A7 | Bracket FEA

## Design 1

Mesh 1: Course
- SF = .216
- $\sigma_{max}$ = 185.82 ksi
- $\delta_{max}$ = .136 in
- mass = .268 lb

Mesh 2: finer/course
- SF = .189
- $\sigma_{max}$ = 211.35 ksi
- $\delta_{max}$ = .136 in
- mass = .268 lb

Mesh 3: Fine
- SF = .180
- $\sigma_{max}$ = 221.76 ksi
- $\delta_{max}$ = .136 in
- mass = .268 lb

Convergence: Mesh 1 & 2

$SF = \frac{|.189 - .216|}{.189} \times 100 = 14.29\%$ X

$\sigma_{max} = \frac{|211.35 - 185.82|}{211.35} \times 100 = 12.08\%$ X

$\delta_{max} = \frac{|.136 - .136|}{.136} \times 100 = 0\%$ ✓

$mass = \frac{|.268 - .268|}{.268} \times 100 = 0\%$ ✓

Convergence: Mesh 2 & 3

$SF = \frac{|.180 - .189|}{.180} \times 100 = 5\%$ ✓

$\sigma_{max} = \frac{|221.76 - 211.35|}{221.76} \times 100 = 4.6\%$ ✓

$\delta_{max} = 0\%$ ✓

$mass = 0\%$ ✓

## Design 2

Mesh 1: course
- SF = .244
- $\sigma_{max}$ = 163.66 ksi
- $\delta_{max}$ = .129 in
- mass = .268 lb

Mesh 2: finer/course
- SF = .226
- $\sigma_{max}$ = 176.4 lb
- $\delta_{max}$ = .129 in
- mass = .268 lb

Mesh 3: Fine
- SF = .241
- $\sigma_{max}$ = 165.74 lb
- $\delta_{max}$ = .130 in
- mass = .268 lb

Convergence: Mesh 1 & 2

$SF = \frac{|.226 - .244|}{.226} \times 100 = 7.96\%$ X

$\sigma_{max} = \frac{|176.4 - 163.66|}{176.4} \times 100 = 7.22\%$ X

$\delta_{max} = 0\%$ ✓

mass = 0% ✓

Convergence: Mesh 2 & 3

$SF = \frac{|.241 - .226|}{.241} \times 100 = 6.22\%$ ✓

$\sigma_{max} = \frac{|165.74 - 176.4|}{165.74} \times 100 = 6.43\%$ ✓

$\delta_{max} = \frac{|.130 - .129|}{.130} \times 100 = .77\%$ ✓

mass = 0% ✓

## Design 3

Mesh 1: course
- SF = 2.86
- $\sigma_{max}$ = 13.96 ksi
- $\delta_{max}$ = .0039 in
- mass = .2931b

Mesh 2: finer/course
- SF = 2.83
- $\sigma_{max}$ = 14.11
- $\delta_{max}$ = .0039 in
- mass = .2931b

Convergence: Mesh 1 & 2

$SF = \frac{|2.83 - 2.86|}{2.83} \times 100 = 1.06\%$ ✓

$\sigma_{max} = \frac{|14.11 - 13.96|}{14.11} \times 100 = 1.06\%$ ✓

$\delta_{max} = 0\%$

mass = 0%